

GCI WORLD FINAL ASSIGNMENT

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BUSINESS OVERVIEW/CONTEXT

Market Analysis

Telecommunications is one of the most data-intensive industries in the world ,generating vast amounts of data on customers ,network performance , operations, and more(<https://www.thoughtspot.com/data-trends/analytics/telecom-analytics>). However by leveraging the power of data telecom companies can stay ahead of the curve and drive for their customers and shareholders .Its essence lies in harnessing extensive datasets to extract valuable insights, enabling more informed decision through an enhanced understanding of various options. Privacy breaches, misleading marketing, and poor customer service are among the biggest challenges in the telecommunications industry — and remain the most significant threats to the sector's reputation, according to our latest insights report(<https://www.groupcaliber.com/bad-signals-the-telecom-sectors-reputation-problem-and-how-to-fix-it/>). In other words, to improve their reputation, telecom companies need to acknowledge and address concerns about data breaches and deceptive advertising, improve customer service, provide more reliable coverage and introduce much more transparent pricing.

This document summarizes building a data solution for a telecom company A. The goal is to run an Exploratory Data Analysis (EDA) on the provided datasets ,identify core business problems and create data driven solutions with quantifiable and evidence based impacts.

DATA OVERVIEW

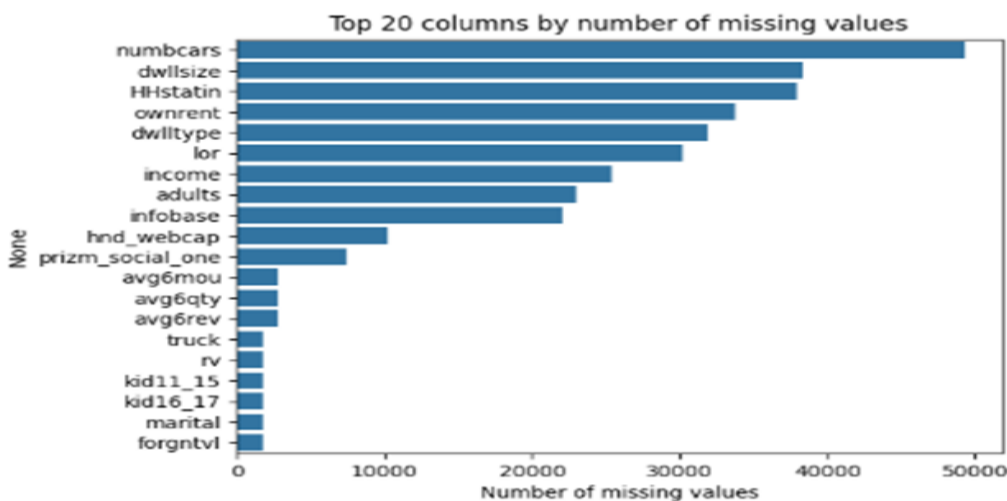
ABOUT THE DATASET

The dataset provided by Company A contains two types of .csv files with customer information (Client) and usage history (Record). It is a large-scale table data of approximately 100,000 records with 100 columns after merging.

Client: (100000, 50) Merged: (100000, 100)
Record: (100000, 51)

DATA OVERVIEW

The dataset contained a substantial number of missing values in some of the columns. Columns with missing values greater than 30% were removed and the remaining ones were imputed.

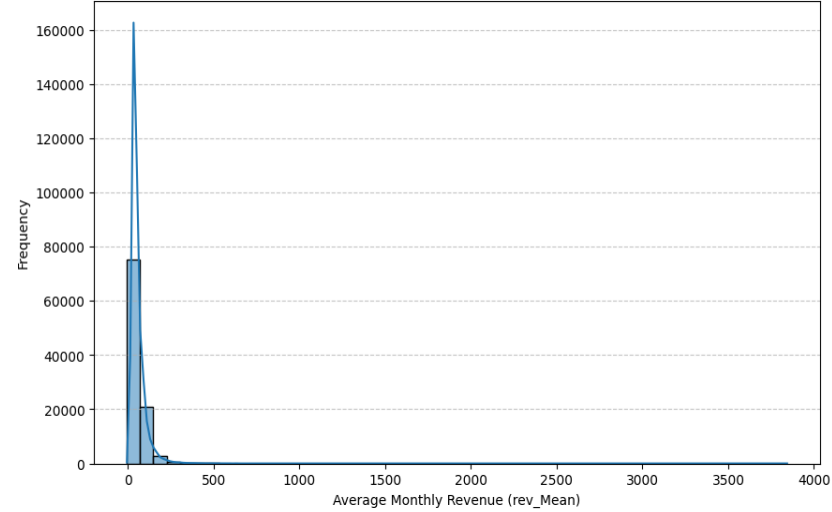


Top 20 % of columns with missing values

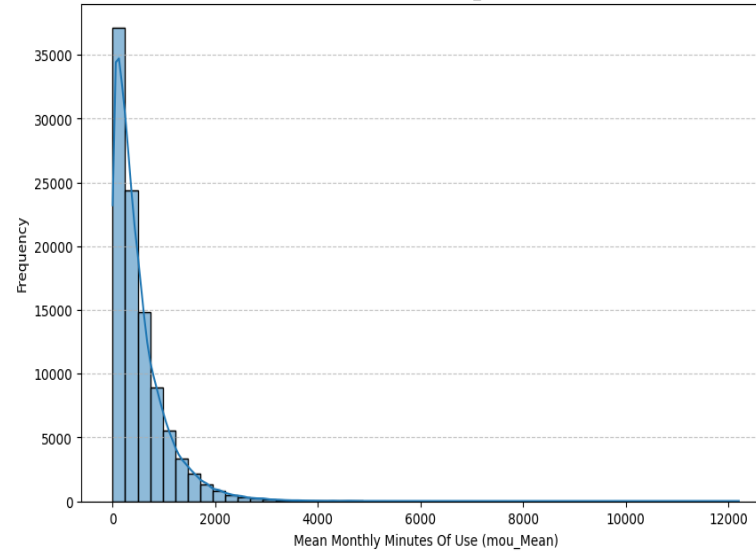
numbcars	49.366
dwllsize	38.308
HHstatin	37.923
ownrent	33.706
dwlltype	31.909
lor	30.190
income	25.436
adults	23.019
infobase	22.079
hnd_webcap	10.189
prizm_social_one	7.388
avg6mou	2.839
avg6qty	2.839
avg6rev	2.839
truck	1.732
rv	1.732
kid11_15	1.732
kid16_17	1.732
marital	1.732
forgntvl	1.732

EXPLORATORY DATA ANALYSIS

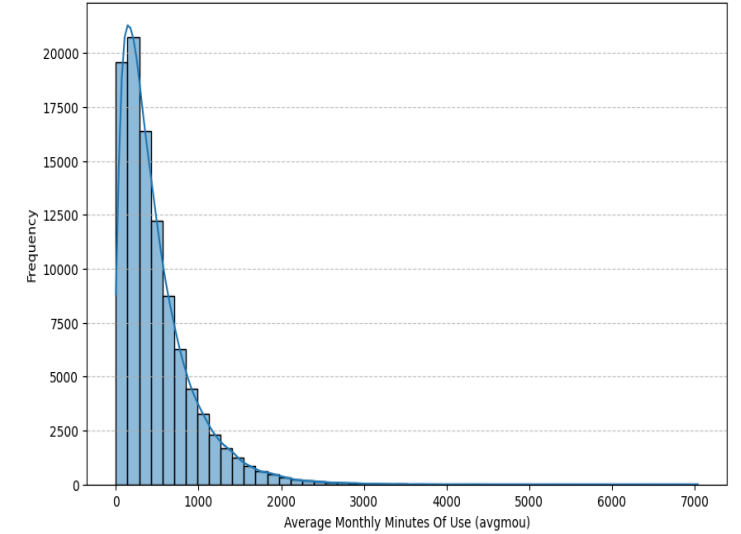
Distribution of rev_Mean



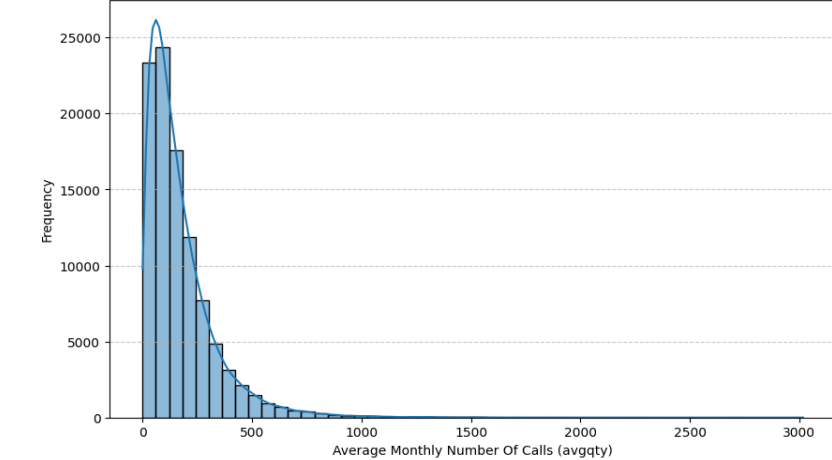
Distribution of mou_Mean



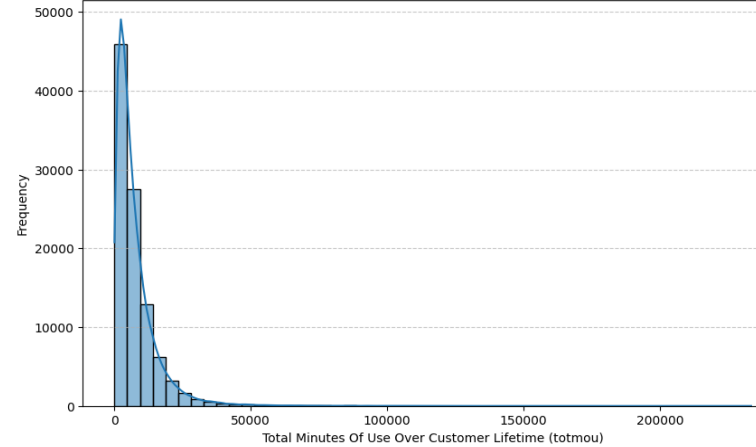
Distribution of avgmou_Mean



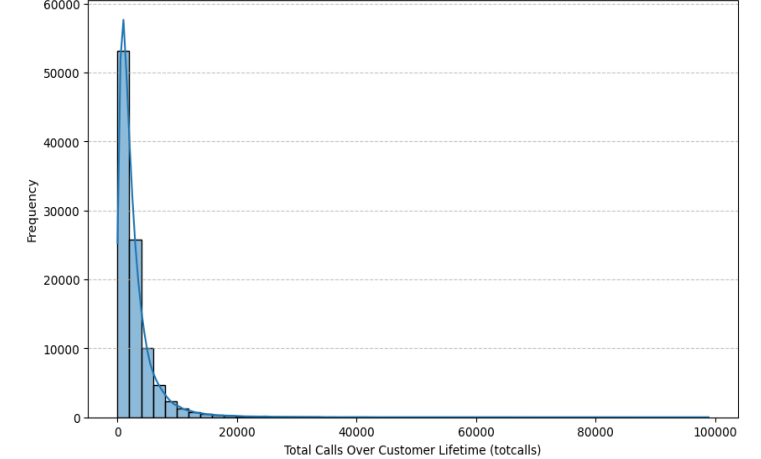
Distribution of avgqty



Distribution of totmou

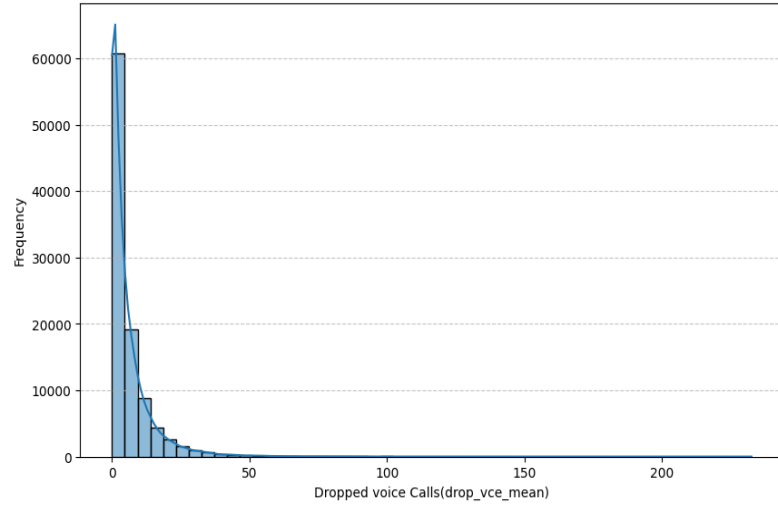


Distribution of totcalls

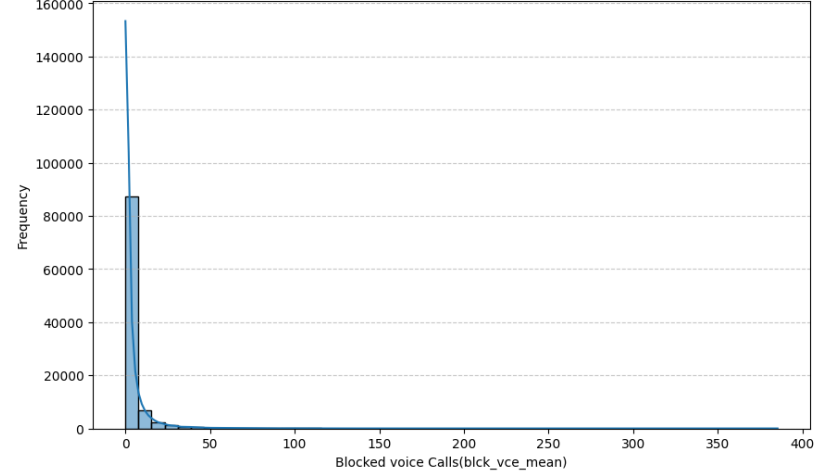


EXPLORATORY DATA ANALYSIS

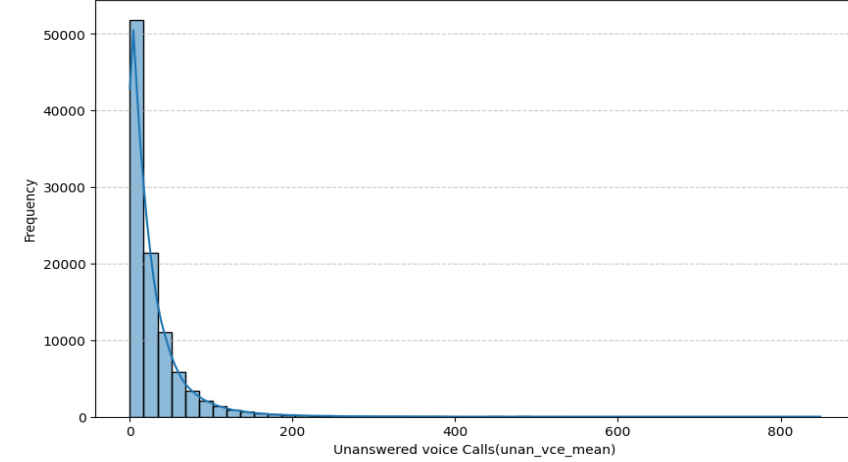
Distribution of Dropped Voice Calls



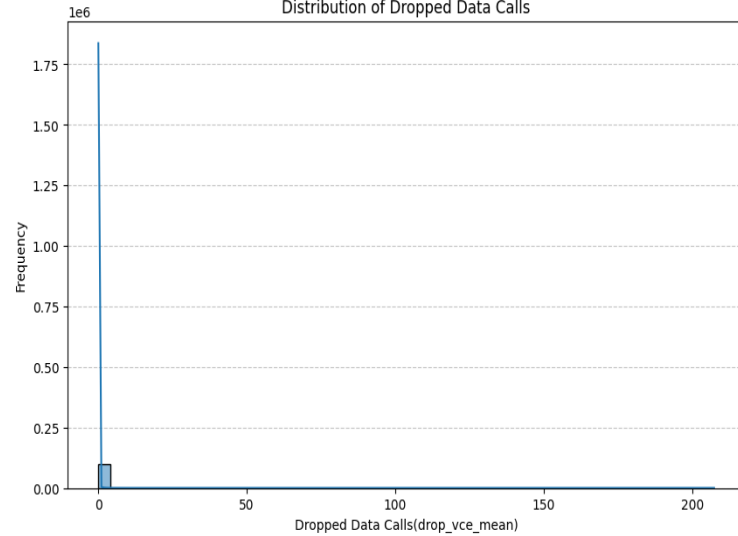
Distribution of Blocked Voice Calls



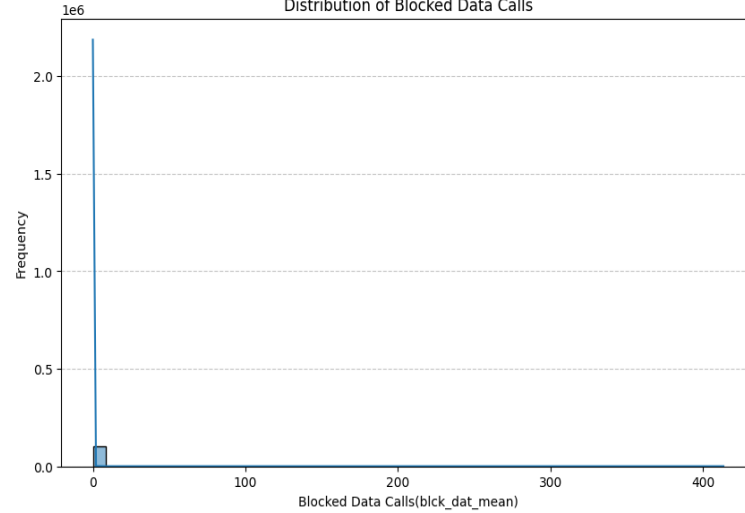
Distribution of Unanswered Voice Calls



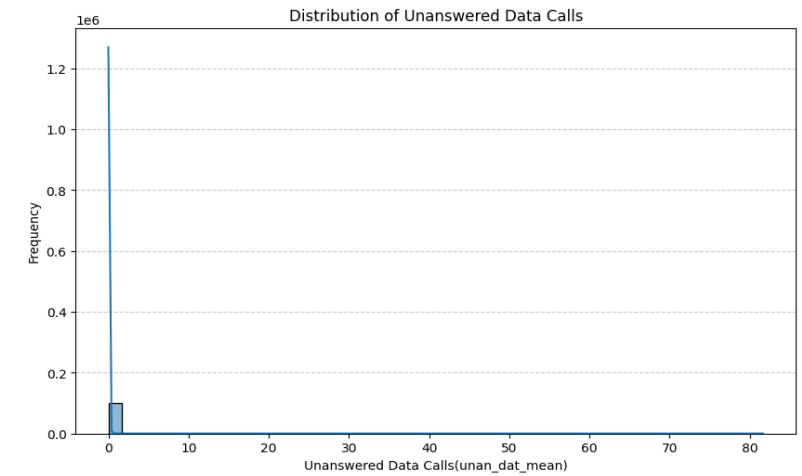
Distribution of Dropped Data Calls



Distribution of Blocked Data Calls

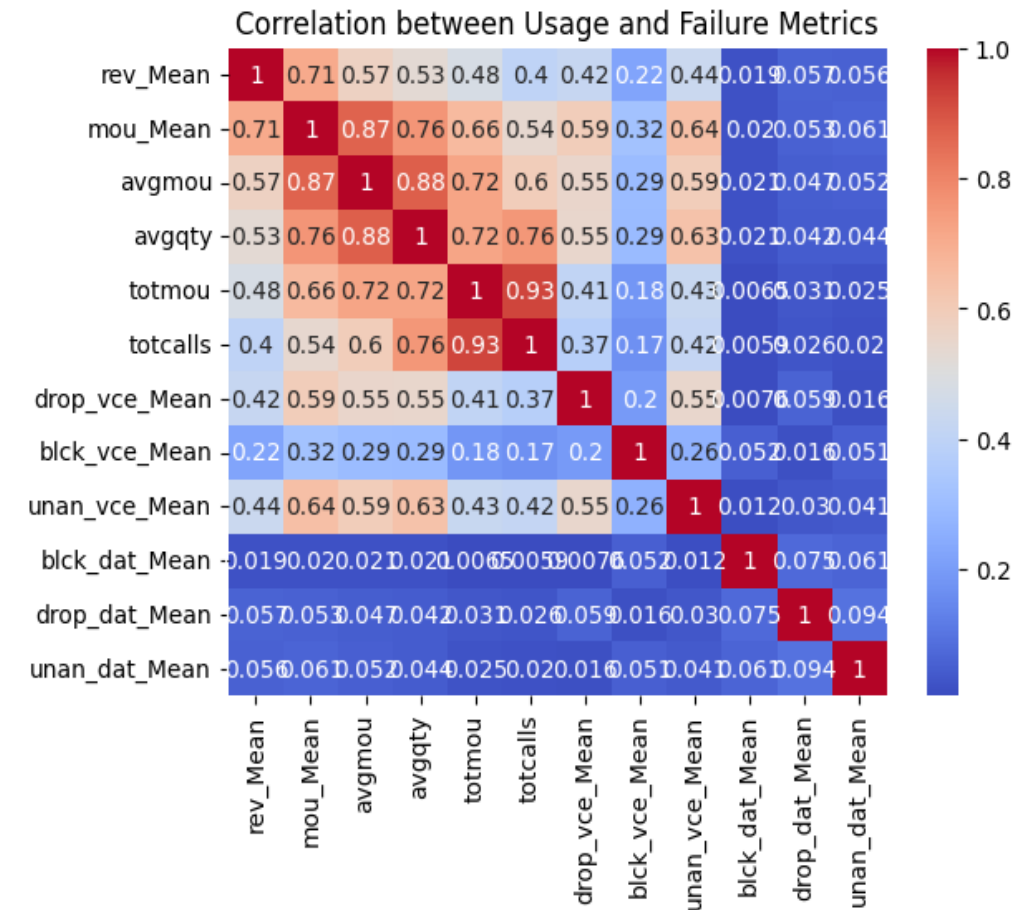


Distribution of Unanswered Data Calls



EXPLORATORY DATA ANALYSIS

The heatmap visualizes the correlations between various usage and failure metrics in the dataset. The color intensity and hue indicate the strength and direction of the relationships, with red representing strong positive correlations and blue representing strong negative correlations.

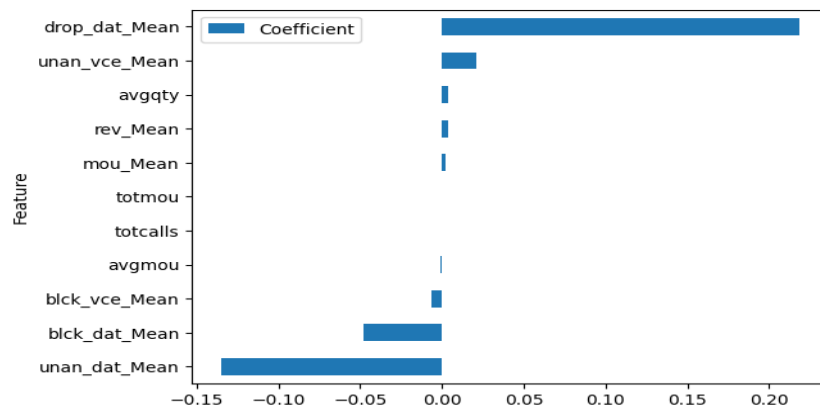


Feature Engineering

To predict customers likely to experience high dropped-call rates, relevant customer usage and network performance variables were selected:

- ❑ Revenue (rev_Mean)
- ❑ Minutes of Use (mou_Mean, avgmou, totmou)
- ❑ Call Volume (avgqty, totcalls)
- ❑ Voice Service Quality Metrics (blk_vce_Mean, unan_vce_Mean)
- ❑ Data Service Quality Metrics (blk_dat_Mean, drop_dat_Mean, unan_dat_Mean)

The target variable High_drop was created by classifying customers whose dropped voice call rate exceeded the dataset average.



The model predicts customers who are likely to experience poor network quality before service issues become severe. This enables proactive network optimization, targeted maintenance, and improved customer satisfaction.

Model Selection

Selected Model: Logistic Regression

Logistic Regression was chosen because it is:

- ❑ Suitable for binary classification problems
- ❑ Provides interpretable results for business stakeholders
- ❑ Efficient for large telecom datasets
- ❑ Produces probability scores that can be used to rank customer risk levels

Evaluation Metrics

- ❑ Accuracy
- ❑ ROC-AUC Score

MODEL PERFORMANCE

A Logistic Regression model was developed to predict customers likely to experience above-average dropped voice calls (High_drop).

The dataset was split into:

80% Training Data

20% Testing Data

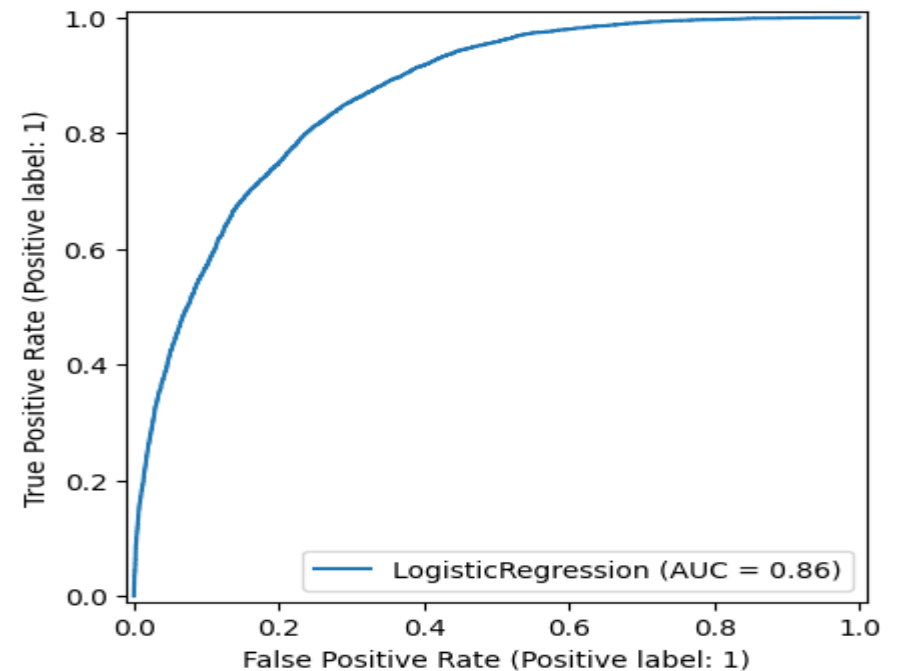
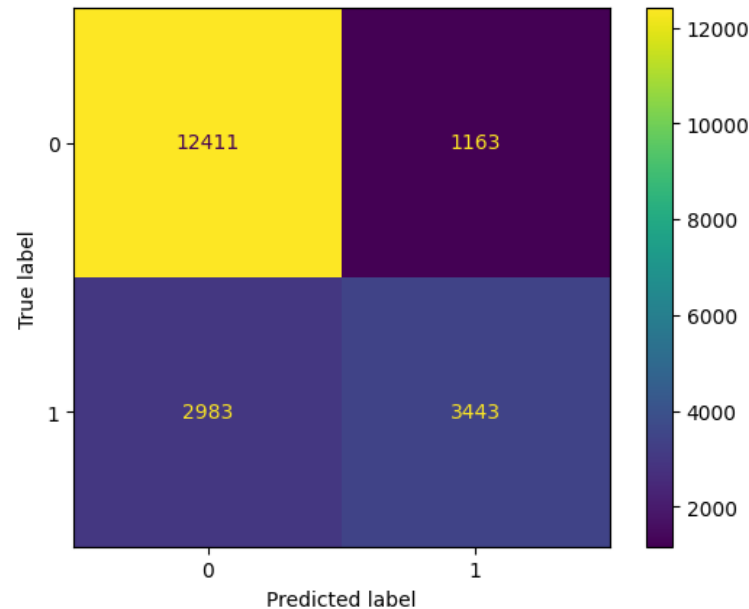
	precision	recall	f1-score	support
0	0.81	0.91	0.86	13574
1	0.75	0.54	0.62	6426
accuracy			0.79	20000
macro avg	0.78	0.73	0.74	20000
weighted avg	0.79	0.79	0.78	20000
Accuracy: 0.7927				
ROC AUC: 0.8649566271837222				

Interpretation

The model correctly classified approximately 79% of customers into high-risk and low-risk dropped-call categories.

ROC-AUC

The ROC-AUC score of 86% indicates the model's ability to distinguish between customers likely and unlikely to experience excessive dropped calls.



The Logistic Regression model successfully predicts customers at risk of experiencing excessive dropped calls, enabling proactive network management and improved service quality.

Business problem

Telecommunication companies rely on high-quality network performance to maintain customer satisfaction and revenue. Analysis of the customer usage and network performance data revealed that a significant number of customers experience above-average dropped voice calls, indicating potential service quality issues.

Frequent dropped calls negatively affect customer experience, increase customer complaints, reduce service usage, and may ultimately lead to customer churn. Currently, network issues are often addressed reactively after customers have already been affected.

Business Objective

To develop a data-driven solution that can identify customers at high risk of experiencing excessive dropped voice calls before service quality deteriorates.

Expected Business Value

- ☐ Improve Quality of Service (QoS)
- ☐ Enable proactive network maintenance
- ☐ Reduce customer complaints
- ☐ Increase customer satisfaction and retention
- ☐ Support data-driven infrastructure investment decisions

Business Proposal

Based on the analysis, we propose implementing a machine learning-driven Network Quality Monitoring System that predicts customers at risk of experiencing excessive dropped voice calls.

The system will continuously analyze customer usage patterns and network performance indicators to identify high-risk customers before service quality issues become severe

Business Impact

By identifying high-risk customers before service quality deteriorates, the company can:

- ☐ Prioritize network maintenance
- ☐ Improve Quality of Service (QoS)
- ☐ Reduce customer complaints
- ☐ Increase customer satisfaction
- ☐ Support proactive operational decision-making

Proposed Solution

☐ **Customer Experience Improvement**

Reduce dropped-call incidents.

Improve Quality of Service (QoS).

Decrease customer complaints and service disruptions.

☐ **Predict Network Quality Issues**

Use the Logistic Regression model to identify customers with a high probability of experiencing dropped calls.

Generate risk scores for individual customers and network segments.

☐ **Proactive Network Optimization**

Prioritize maintenance and troubleshooting in high-risk areas.

Allocate network resources more efficiently.

Detect emerging service quality issues before widespread customer impact.

Expected Business Benefits

☐ Improved network reliability

☐ Enhanced customer satisfaction

☐ Reduced operational costs through targeted maintenance

☐ Better utilization of network infrastructure

☐ Data-driven decision-making for future network investments

Success Metrics

- ☐ Reduction in dropped-call rates
- ☐ Reduction in customer complaints
- ☐ Improvement in customer satisfaction scores
- ☐ Increased network service availability
- ☐ Improved operational efficiency

Projected Impact

- ☐ Early detection of network quality issues
- ☐ More efficient maintenance planning
- ☐ Improved customer retention through better service quality
- ☐ Stronger competitive position in the telecommunications market

Quantified Business Impact (Based on Assumptions)

Assuming a total customer base of 100,000 subscribers with an Average Revenue Per User (ARPU) of \$10 per month, and that 15% of customers are affected by network quality issues, the deployment of the predictive network quality monitoring system is projected to generate significant financial and operational benefits

Monthly Revenue at Risk:
(This represents the monthly revenue impacted by poor network performance.)

$$100,000 * 15\% * \$10 = \$150,000$$

Annual Revenue Impact (if issues persist without intervention):

$$\$150,000 * 12 = \$1,800,000$$

Estimated Reduction in Customer Churn

With improved network reliability, the churn rate due to poor service (currently at 5%) could decrease by approximately 20-25%, saving about \$90,000 to \$112,500 per month in revenue from retained customers. Cost

Savings from Proactive Maintenance

By targeting maintenance efforts in high-risk areas, operational costs related to network troubleshooting and repairs could decrease by roughly 10-15% of current costs (assuming a proportional cost structure), translating into an estimated annual saving of \$180,000 to \$270,000.

Customer Satisfaction and Retention:

The improved quality of service is expected to reduce complaints by 25-30%, leading to higher customer satisfaction, increased loyalty, and potential upselling opportunities.

Overall Financial Benefit:

Combining revenue retention and operational savings, the company could realize an estimated annual benefit of approximately \$1.5 million to \$2 million from deploying the predictive system.

By leveraging the predictive model, the telecom company can proactively address network issues affecting 15,000 customers (15% of 100,000), reducing customer churn, enhancing satisfaction, and saving operational costs. This translates into a potential annual financial benefit of up to \$2 million, demonstrating the high value of data-driven network management.

REFERENCES

<https://www.groupcaliber.com/bad-signals-the-telecom-sectors-reputation-problem-and-how-to-fix-it/>

<https://www.thoughtspot.com/data-trends/analytics/telecom-analytics>

<https://chatgpt.com/share/6a33f804-7994-83ea-9456-298c1ba34109>